

Benchmark report — DockQ vs. tool confidence

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45 predictions scored across 5 methods and 14 of 14 included targets.

rank_score = each tool's own headline confidence for the top pose (AF3/Protenix ranking_score, Chai aggregate_score, AlphaFold-Multimer iptm+ptm, Boltz confidence_score).
DockQ: incorrect <0.23, acceptable 0.23–0.49, medium 0.49–0.80, high ≥0.80.

1. Results by target

4E81 — Apidaecin fragment · peptide 10 aa, resolved 9/10 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.904	high	0.672	0.927	0.92	0.9	0.89
afmultimer	0.930	high	0.631	0.976	0.8569	-	-
chai	0.946	high	0.551	0.976	0.7338	0.879	0.6975
protenix	0.898	high	0.715	0.902	0.812	0.9194	0.7852

4EZQ — Pyrrhocoricin · peptide 9 aa, resolved 7/9 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
afmultimer	0.587	medium	1.244	0.333	0.8424	-	-
chai	0.210	incorrect	4.226	0.244	0.6849	0.8714	0.6382
protenix	0.793	medium	1.068	0.844	0.8493	0.9232	0.8309

4EZR — Drosocin · peptide 8 aa, resolved 8/8 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.934	high	0.476	0.918	0.95	0.91	0.94
chai	0.919	high	0.487	0.878	0.6296	0.876	0.568
protenix	0.909	high	0.55	0.878	0.9532	0.93	0.959

4EZS — Metchnikowin (Drosophila) · peptide 8 aa, resolved 8/8 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.287	acceptable	3.009	0.184	0.79	0.89	0.74

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
afmultimer	0.294	acceptable	2.735	0.184	0.7	-	-
chai	0.375	acceptable	2.321	0.265	0.5157	0.8746	0.4259
protenix	0.925	high	0.664	0.959	0.9339	0.9314	0.9345

6HY2 — · peptide 12 aa, resolved 12/12 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.289	acceptable	3.275	0.125	0.74	0.94	0.66
afmultimer	0.395	acceptable	2.494	0.25	0.5337	-	-
boltz	0.928	high	0.507	0.9	0.96	0.97	0.88
chai	0.206	incorrect	4.578	0.2	0.3131	0.9212	0.1611
protenix	0.685	medium	1.464	0.7	0.8606	0.9646	0.8346

3QRX — Melittin (different host-protein · peptide 26 aa, resolved 20/26 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.525	medium	2.501	0.786	0.53	0.57	0.36
chai	0.228	incorrect	3.447	0.429	0.4411	0.5948	0.4027
protenix	0.389	acceptable	2.42	0.786	0.6327	0.6233	0.6351

4EZO — PR-39 antibacterial protein · peptide 15 aa, resolved 8/15 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.711	medium	1.486	0.837	0.85	0.89	0.79
chai	0.688	medium	1.554	0.796	0.784	0.8833	0.7592
protenix	0.750	medium	1.302	0.816	0.8536	0.9189	0.8373

4EZU — 'PR-bombesin' from Bombina maxim · peptide 17 aa, resolved 14/17 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.084	incorrect	6.39	0.0	0.7	0.88	0.6
chai	0.090	incorrect	8.43	0.117	0.5948	0.8504	0.5309
protenix	0.178	incorrect	4.557	0.117	0.8298	0.9177	0.8079

6Z2P — Glycodrosocin · peptide 19 aa, resolved 10/19 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.142	incorrect	4.799	0.056	0.54	0.91	0.39

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
afmultimer	0.171	incorrect	4.887	0.167	-	-	-
chai	0.070	incorrect	7.547	0.056	0.362	0.8947	0.2288
protenix	0.769	medium	1.187	0.833	0.6357	0.9393	0.5599

7SAY — LL-37, the canonical human cathe · peptide 37 aa, resolved 35/37 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
chai	0.068	incorrect	14.435	0.148	0.3256	0.5994	0.2571
protenix	0.014	incorrect	18.272	0.0	0.4405	0.5858	0.4042

8AHT — Melittin (bee venom AMP) · peptide 26 aa, resolved 25/26 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.404	acceptable	4.015	0.379	0.56	0.54	0.46
chai	0.550	medium	2.459	0.495	0.6045	0.7742	0.5621
protenix	0.656	medium	1.849	0.663	0.6521	0.7183	0.6355

4JWC — Cathelicidin-3/Bac7-type (Bos ta · peptide 16 aa, resolved 9/16 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
af3	0.672	medium	1.431	0.692	0.83	0.88	0.78
chai	0.685	medium	1.377	0.692	0.4247	0.8379	0.3214
protenix	0.653	medium	1.715	0.692	0.8372	0.9225	0.8159

4JWD — Cathelicidin-3 fragment (Bos tau · peptide 14 aa, resolved 7/14 · quality: caution

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
chai	-	ERROR	-	-	0.5586	0.8601	0.4832
protenix	-	ERROR	-	-	0.7333	0.9071	0.6898

8GAL — Thanatin · peptide 20 aa, resolved 18/20 · quality: ok

method	DockQ	class	iRMSD	fnat	rank_score	ptm	iptm
afmultimer	0.746	medium	0.88	0.808	-	-	-
chai	0.786	medium	0.768	0.827	0.8176	0.8868	0.8003
protenix	0.827	high	0.72	0.923	0.9477	0.9358	0.9507

2. By method

method	n	DockQ mean	≥acceptable	≥medium	≥high	mean rank_score	Spearman(rank_score, DockQ)
af3	10	0.495	8/10	5/10	2/10	0.74	0.71
afmultimer	6	0.520	5/6	3/6	1/6	0.73	0.8
boltz	1	0.928	1/1	1/1	1/1	0.96	n/a
chai	13	0.448	7/13	6/13	2/13	0.56	0.73
protenix	13	0.650	11/13	10/13	4/13	0.79	0.68

3. DockQ matrix (best pose)

target	pep_len	qual	af3	afmultimer	boltz	chai	protenix	best
4E81	10	ok	0.90	0.93	-	0.95	0.90	0.95
4EZQ*	9	caut	-	0.59	-	0.21	0.79	0.79
4EZR	8	ok	0.93	-	-	0.92	0.91	0.93
4EZX	8	ok	0.29	0.29	-	0.38	0.93	0.93
6HY2	12	ok	0.29	0.39	0.93	0.21	0.69	0.93
3QRX*	26	caut	0.53	-	-	0.23	0.39	0.53
4EZO*	15	caut	0.71	-	-	0.69	0.75	0.75
4EZU	17	ok	0.08	-	-	0.09	0.18	0.18
6Z2P*	19	caut	0.14	0.17	-	0.07	0.77	0.77
7SAY	37	ok	-	-	-	0.07	0.01	0.07
8AHT	26	ok	0.40	-	-	0.55	0.66	0.66
4JWC*	16	caut	0.67	-	-	0.69	0.65	0.69
4JWD*	14	caut	-	-	-	ERR	ERR	-
8GAL	20	ok	-	0.75	-	0.79	0.83	0.83

4. Insights

Overall ranking (mean DockQ over each method's scored targets — not the same target set for every method yet, so treat as indicative):

- boltz: 0.928 (n=1)
- protenix: 0.650 (n=13)
- afmultimer: 0.520 (n=6)
- af3: 0.495 (n=10)
- chai: 0.448 (n=13)

Does the tools' own confidence predict DockQ accuracy? Pooled Spearman(rank_score, DockQ) = **0.69** over 41 predictions. Per-method values are in section 2. A high positive value means the tool's self-reported confidence is a usable filter for which predictions to trust.

Hardest targets (lowest *best-of-all-methods* DockQ):

- 7SAY (LL-37, the canonical human cathe): best 0.07, mean 0.04 over 2 methods — quality ok, 35/37 resolved
- 4EZU ('PR-bombesin' from Bombina maxim): best 0.18, mean 0.12 over 3 methods — quality ok, 14/17 resolved
- 3QRX (Melittin (different host-protein): best 0.53, mean 0.38 over 3 methods — quality caution, 20/26 resolved
- 8AHT (Melittin (bee venom AMP)): best 0.66, mean 0.54 over 3 methods — quality ok, 25/26 resolved
- 4JWC (Cathelicidin-3/Bac7-type (Bos ta): best 0.69, mean 0.67 over 3 methods — quality caution, 9/16 resolved
- 4EZO (PR-39 antibacterial protein): best 0.75, mean 0.72 over 3 methods — quality caution, 8/15 resolved

Peptide length vs DockQ: Spearman = **-0.43** over 43 predictions (negative = longer peptides score worse).

DockQ could not score:

- chai/4JWD: ERROR RuntimeError: DockQ produced no JSON (ERROR:root:Could not find interfaces in the native model. Please double check the inputs or select different chains with the --mapping flag.)
- protenix/4JWD: ERROR RuntimeError: DockQ produced no JSON (ERROR:root:Could not find interfaces in the native model. Please double check the inputs or select different chains with the --mapping flag.)

Caveats:

- caution targets (6Z2P, 4JWC, 4JWD, 4EZQ, 4EZO, 3QRX) have a partly-unresolved native peptide, so DockQ scores them over a fragment — weaker evidence, marked * in section 3.
- 4JWD: the 7 resolved peptide residues sit ~6 Å from the receptor (no contact), so DockQ finds no native interface at all — this target's reference is effectively unusable; consider reclassifying it from caution to drop.
- AlphaFold-Multimer confidence is only iptm+ptm (combined); ptm/iptm not reported separately by that pipeline.
- Method means are over different target subsets until every method has run on all included targets.